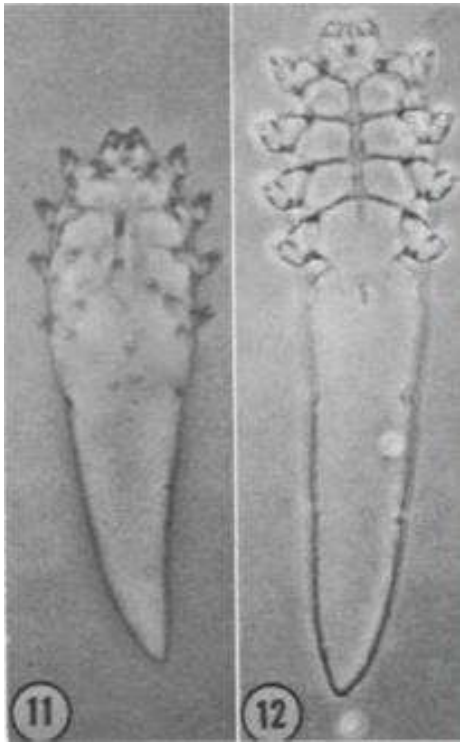
PARASITIC LID DISEASE

Lorne Yudcovitch, O.D., M.S., F.A.A.O.

DEMODEX INFESTATION/DEMODICOSIS aka MITES

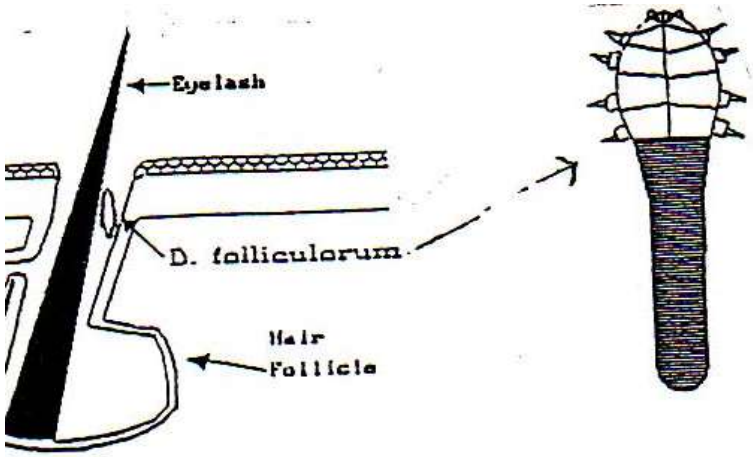
- Eyelid (also facial hair and eyebrow) mite infestation
- Prevalence
  - 0-25 years: 29%
  - 26-50 years: 53%
  - 51-90 years: 67% → ubiquitous
- Contributing factors
  - Mother-infant transmission (nursing)
  - Compromised immune system
  - Vector theory of spreading staph infection?



Demodex infestation/Demodicosis

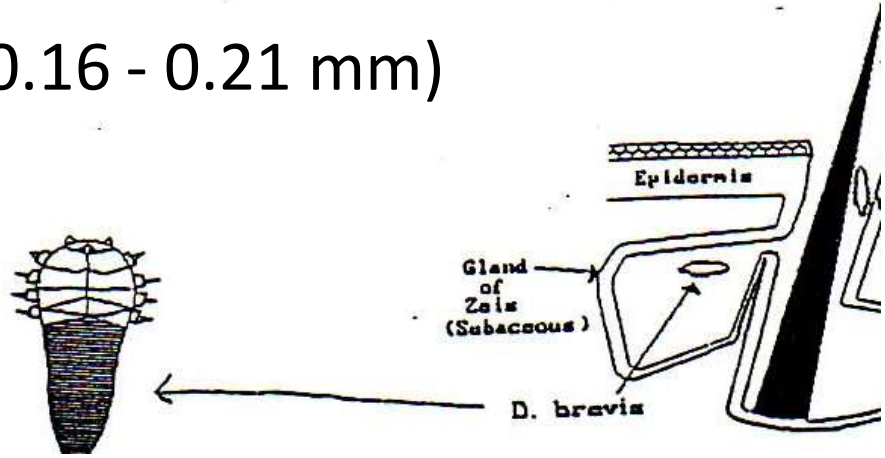
Demodex folliculorum

- Cigar shaped (0.28 - 0.44 mm)
- Colonial (3+ per follicle)
- Upper section of hair follicle
- Collarettes/cuffs
  - Best clinical diagnostic sign

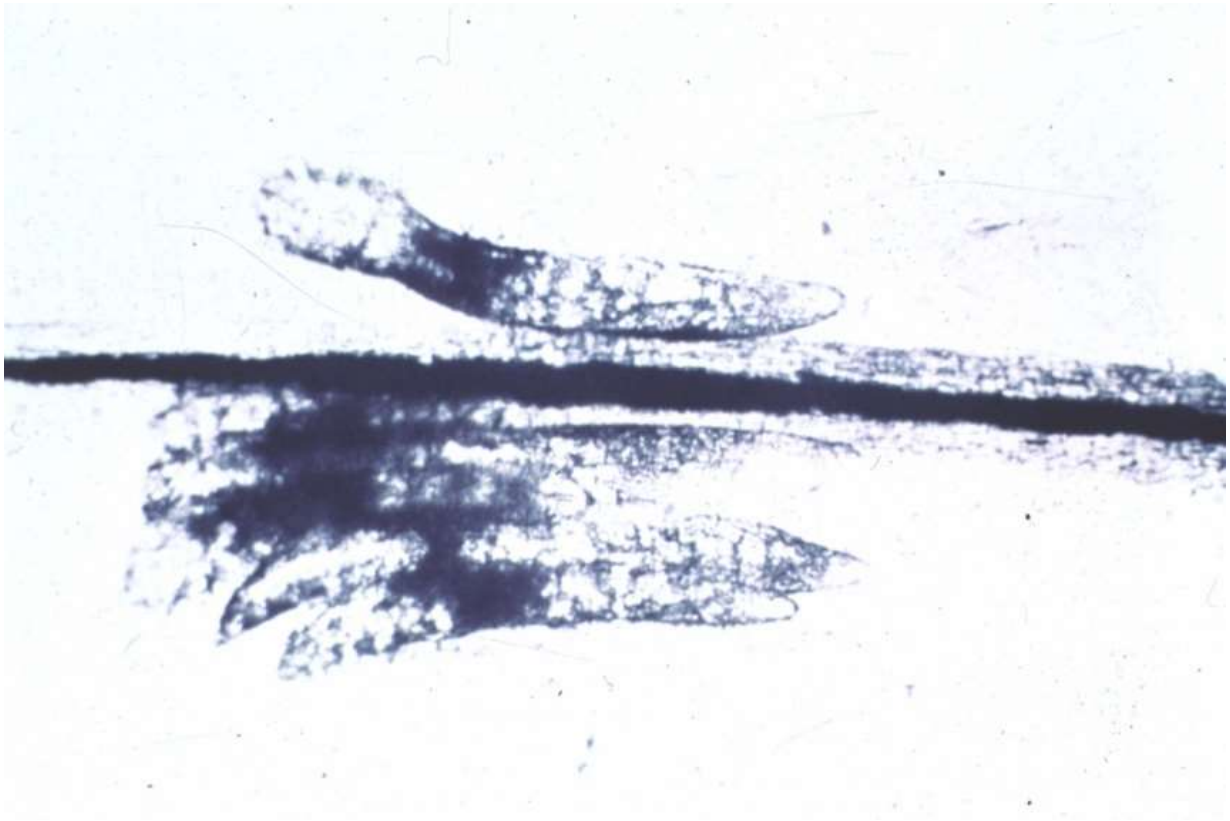


Demodex brevis

- Shorter, stubbier species (0.16 - 0.21 mm)
- Solitary
- Sebaceous glands
- Damage/plugging of gland
  - Dry eye
  - Chalazion formation



Demodex Folliculorum

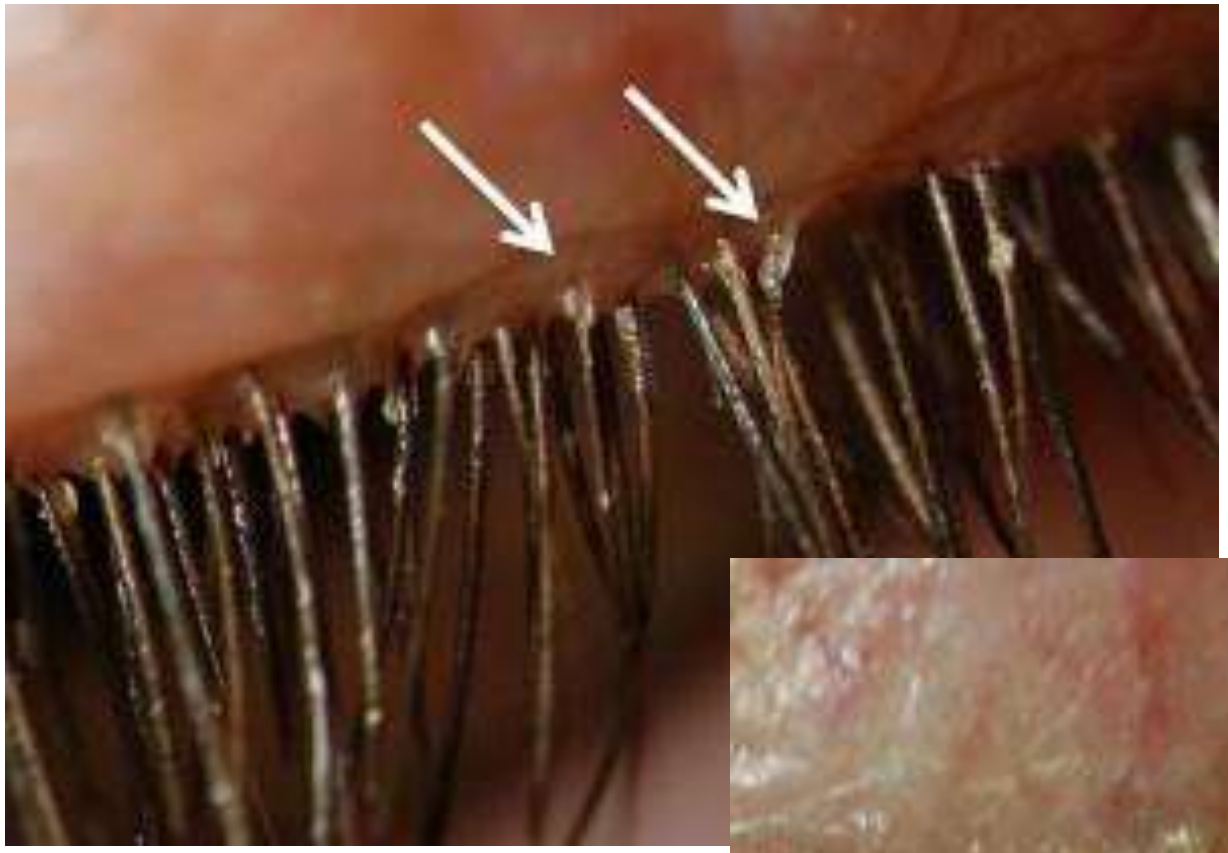


Demodex Folliculorum

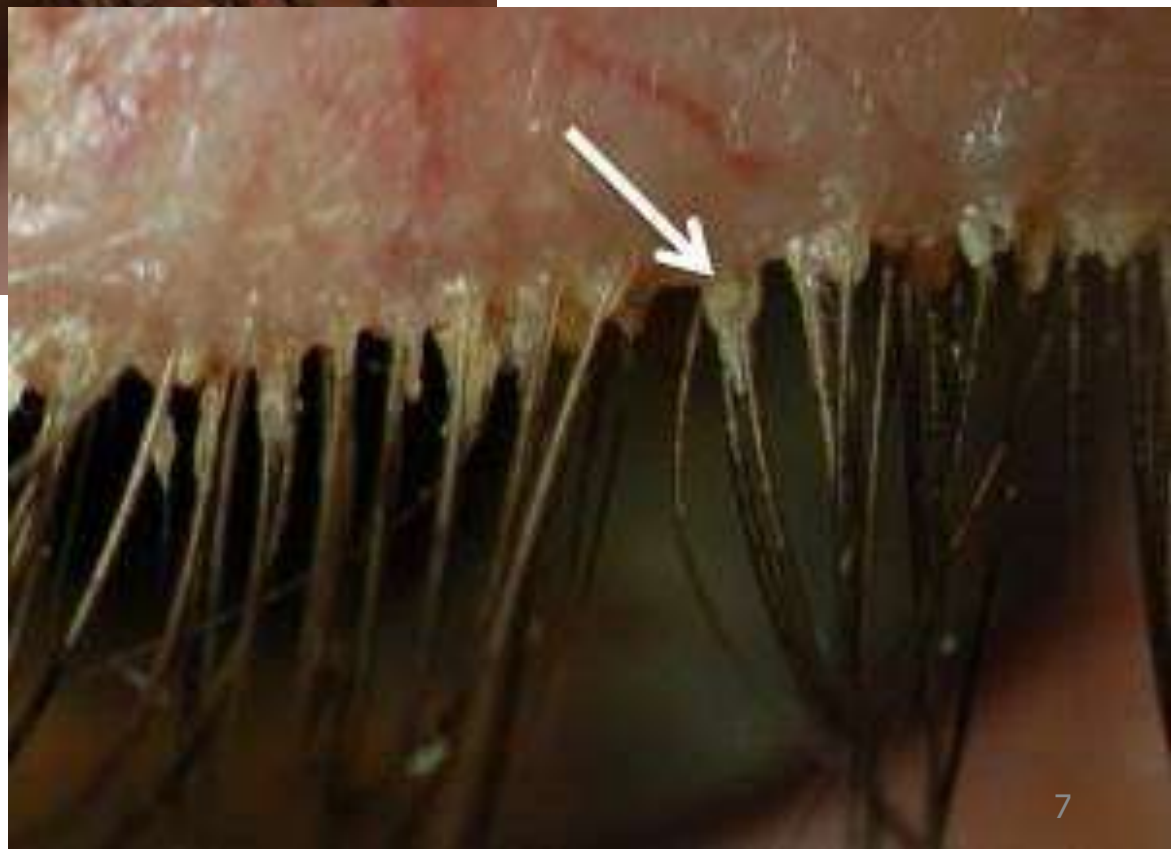


Demodex Lifecycle and Behavior

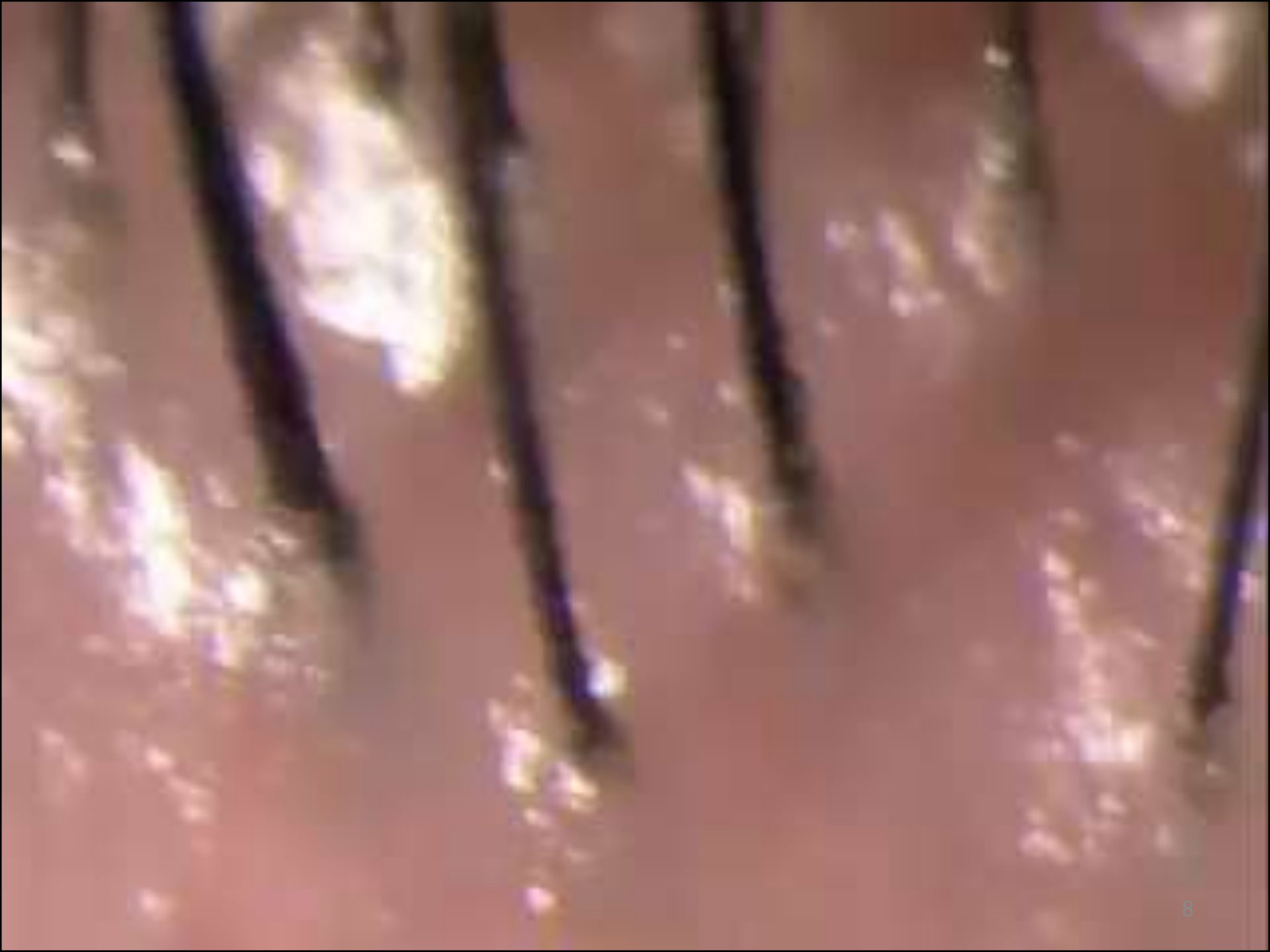
- Life cycle between 14 to 18 days
  - Egg → larva → adult
- Female mites lay up to 10-12 eggs at a time
  - Leads to overpopulation if unmanaged
- *Demodex* avoids light; buries head-first in follicles or glands, with only tail visible at lash follicle opening
  - Lays eggs, ingests gland lining
  - Results in follicular base distension and trichiasis
- Debris generated collects at lash roots
  - Appears as gelatinous cylindrical ‘sleeve’ or cylindrical dandruff (CD) at eyelash base (pathognomonic for mite infestation)



Gelatinous cylindrical dandruff sleeves



Triangular distension of follicle base





https://contactlensupdate.com/2015/12/08/demodex-folliculorum/

Demodex infestation/Demodicosis

Subjective

- Itching – especially in morning
- Burning
- Crusting
- Swelling of lid margins
- Lash loss



Objective

- Lid margin erythema
- Conj. injection
- Blepharitis
- Cuffing around lashes
- Plugging of meibomian glands
- Lash loss = _____

Demodex Folliculorum



Chalazion



Madarosis



Demodex infestation/Demodicosis



Assessment/DDX

- Case HX
- Patient's age
- Clinical signs

Plan/TX

- In-office procedure – aggressive lash scrubs
- Lid hygiene – Tea tree oil products may work
- Ung hs x 10-14 days (bland & viscous)
- Anti-microbial TX – for any super-infections

Management











Tea tree oil derived from leaves of *Melaleuca alternifolia*, an Australian tree

Cliradex (Bio-Tissue)

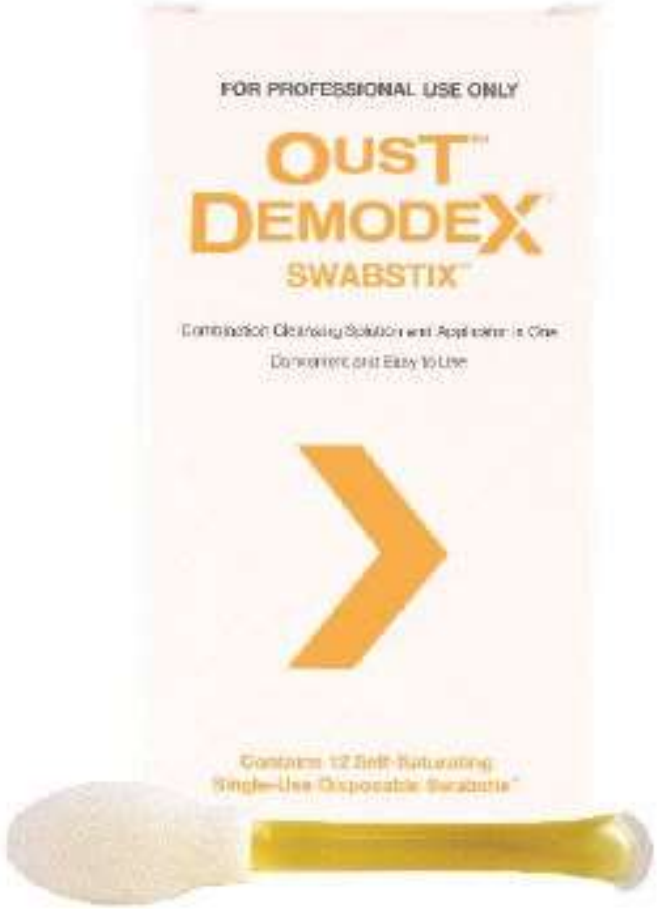
- Natural, preservative-free lid, lash, facial cleanser
- Has specific tea tree oil component (4-Terpineol)
  - More potent miticidal than tea tree oil as a whole
  - May reduce adverse/antagonistic effects of other components of tea tree oil
  - Scientifically proven to eliminate Demodex



- Safe to use daily; eliminates signs and symptoms
- Towelette qAM & qHS x 2 wks, then HS till no signs
- Clean eyelids/margins/eyelashes/periocular area
- Keep eyes closed for 1 min while areas air dry

Ocusoft Oust Demodex Swabstix™

"Utilizing a unique blend of all natural oils including tea tree oil, sea buckthorn and coconut oil, Oust™ Demodex® Swabstix™ aid in the removal of adult Demodex mites and their offspring when used on the skin and skin structures surrounding the eye."
(Ocusoft)



Xdemvy



- Tarsus Pharmaceuticals
- lotilaner 0.25% ophth. sol.
- FDA approved July 2023
  - Saturn-1 and Saturn 2 trials
  - About 50% patients had reduction to 2 or less collarettes, and about 60% patients had *Demodex* mites completely eradicated by 6 weeks
- 1 drop both eyes twice daily (q12hr) for 6 weeks
- Sting/burn 10%; punctate keratitis/stye <2%


NOW AVAILABLE
TO PRESCRIBE
Learn more at
XDEMVYHCP.com
xdemvy™
(lotilaner ophthalmic
solution) 0.25%
Might over mites
in just 6 weeks
Real results
BEFORE
AFTER
44% and 55% of patients taking XDEMVY in SATURN-1 (N=209) and SATURN-2 (N=193), respectively, achieved a significant improvement in their eyelids (reduction of collarettes to no more than 2 collarettes per upper lid) at Day 43 vs 7% (N=204) and 12% (N=200) of patients taking vehicle (P<0.01 in each trial).*
All images are of actual patients who participated in clinical trials for Tarsus Pharmaceuticals.

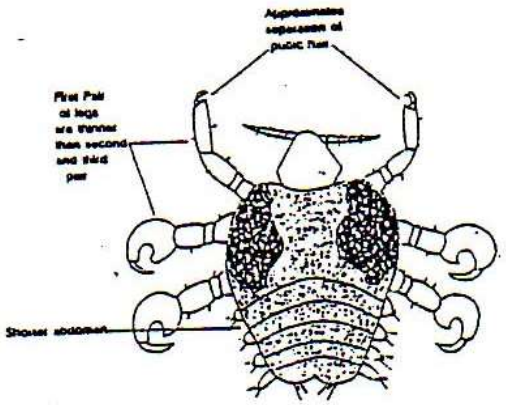
Demodex treatment considerations

- Mites lifespan: 2-3 weeks
- Female will lay about 20 eggs in single lash follicle in that time
- Patients need several weeks (6-8 weeks in some cases) of treatment and maintenance
- Immunocompromised patients may retain more mites and for longer time periods
  - Mites may produce immunologic suppressants

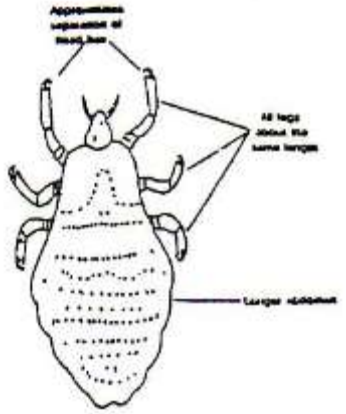
PHTHIRIASIS PALPEBRARUM*/ PEDICULOSIS PALPEBRARUM aka LICE

Eyelid infestation with lice

1. *Phthirus pubis* (crab louse)
2. *Pediculus corporis* (body louse)
3. *Pediculus capitus* (head louse)

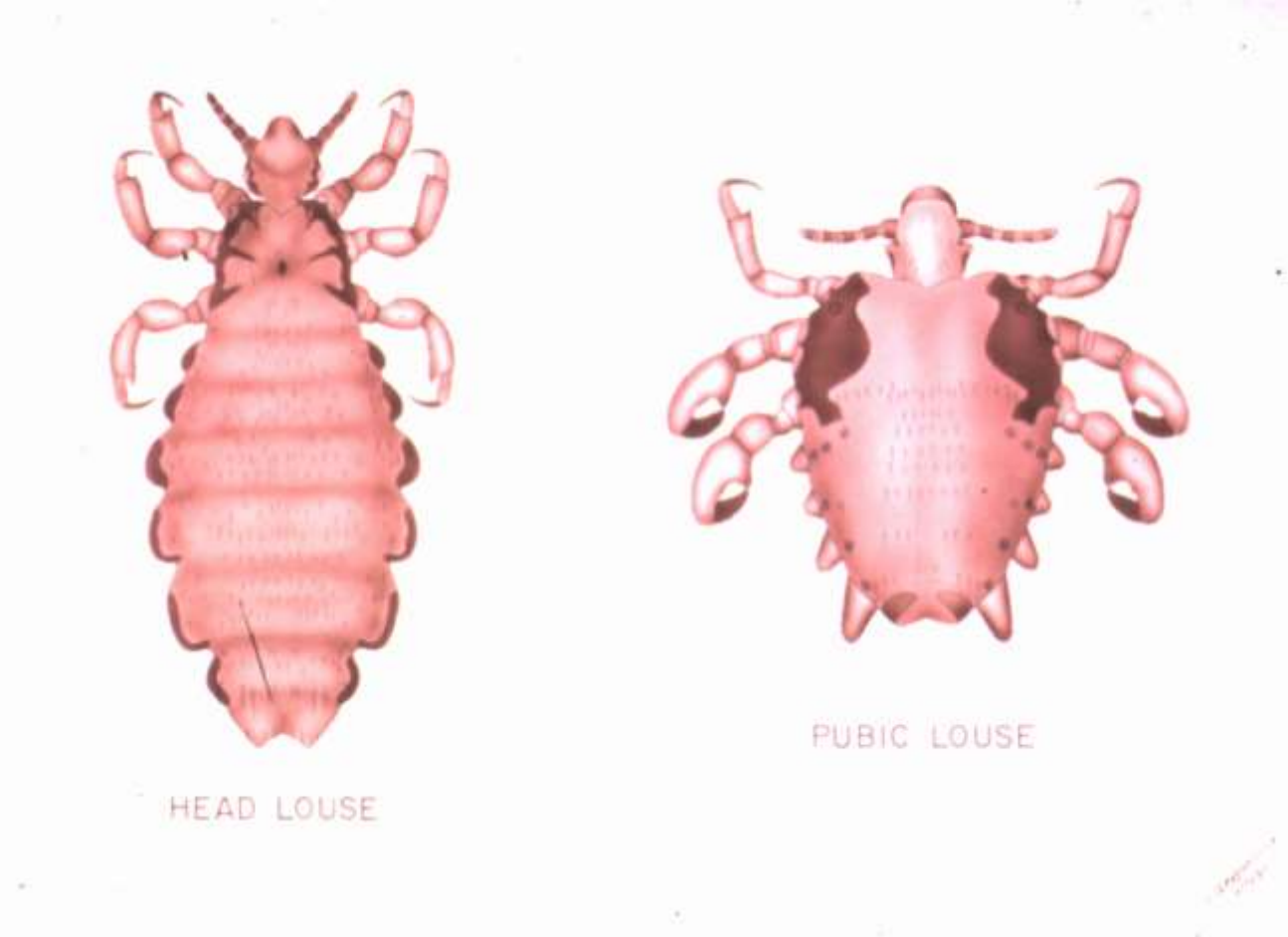


- Vector of serious diseases (typhus)
- Free-moving (*Pediculus*) vs. localized (*Phthirus*)
- Nits (eggs)
- Phthiriasis may be considered a STD

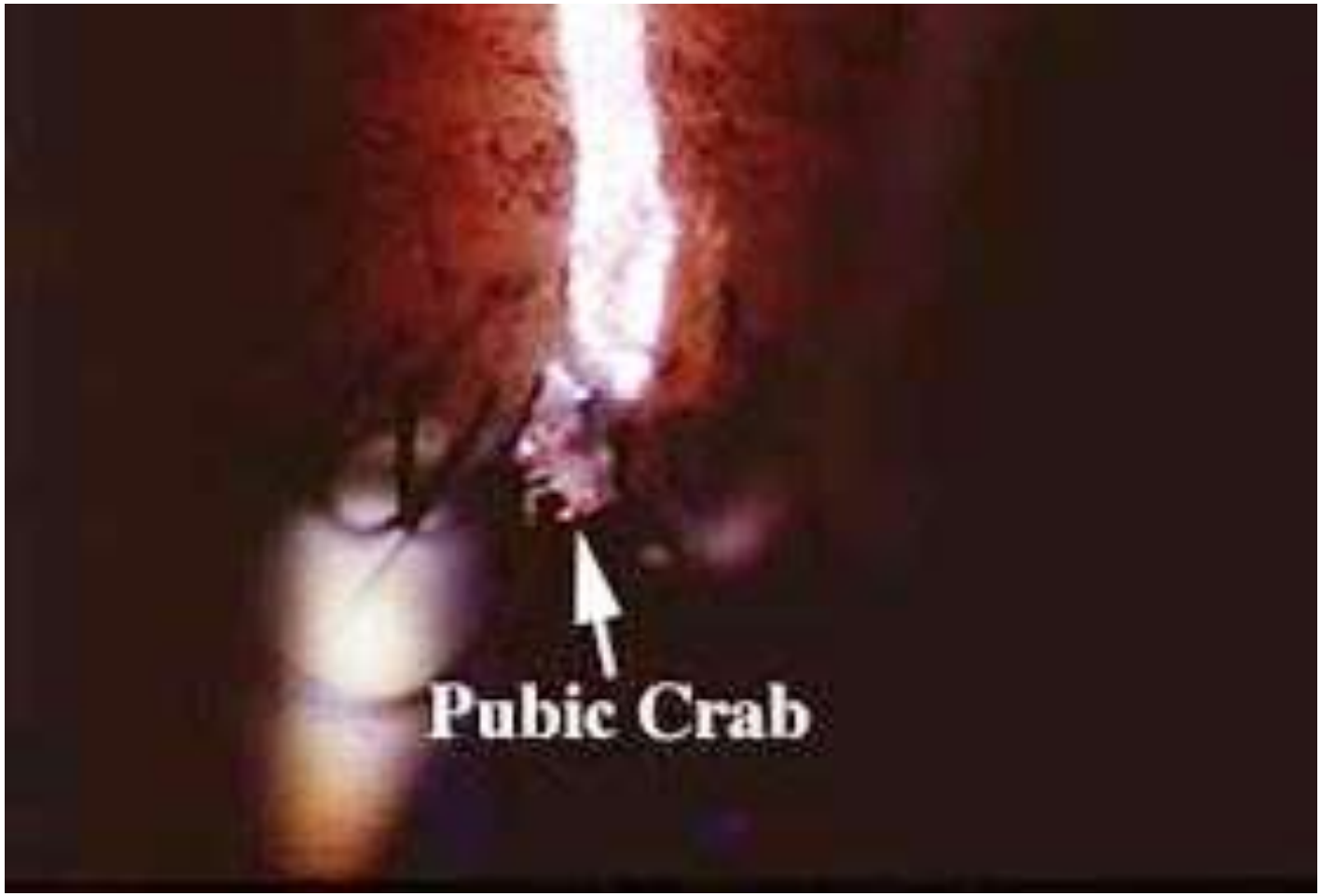


*also called phthiriasis ciliaris or ciliary phthiriasis

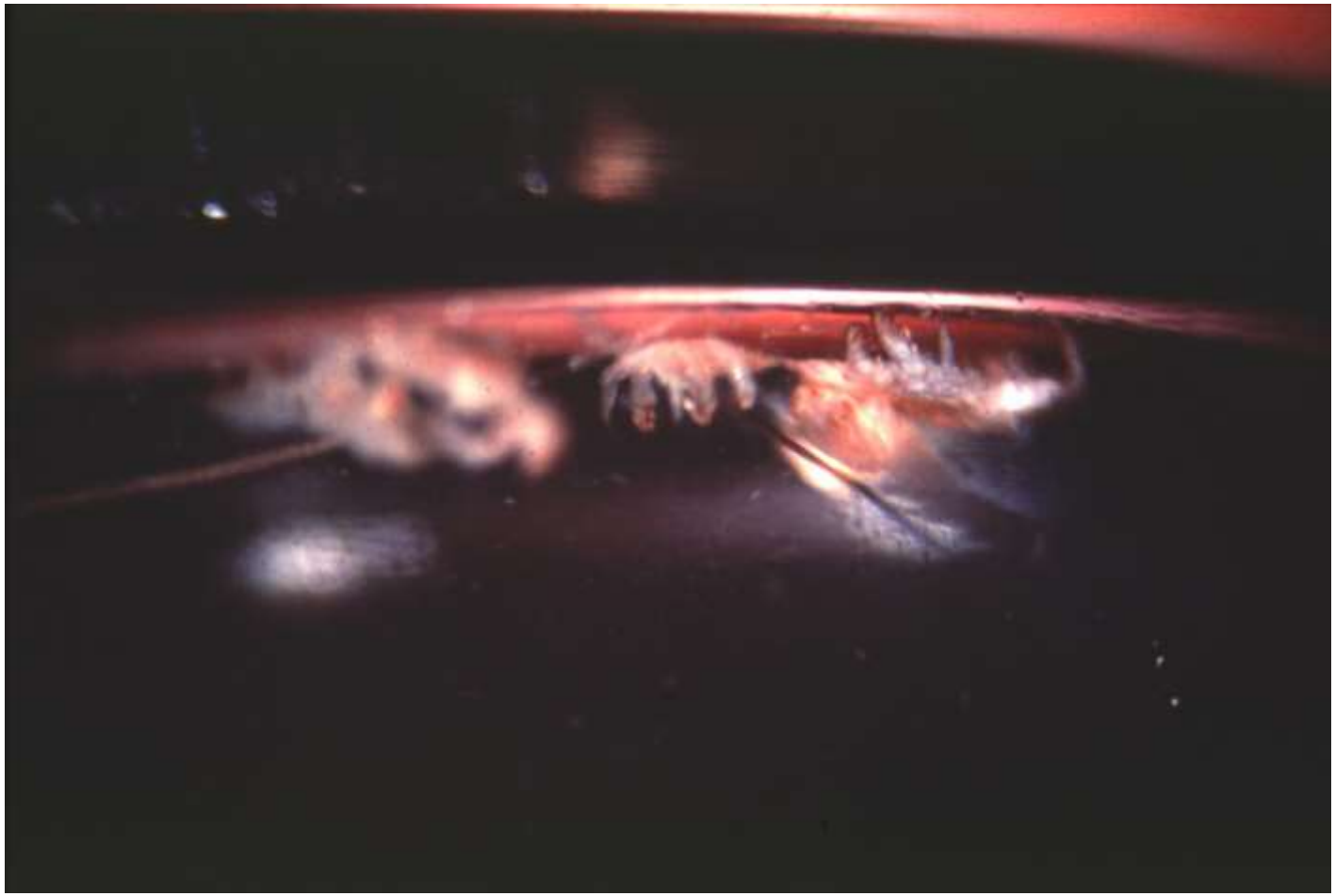
Head & Pubic Lice



Phthiriasis pubis



Phthiriasis pubis

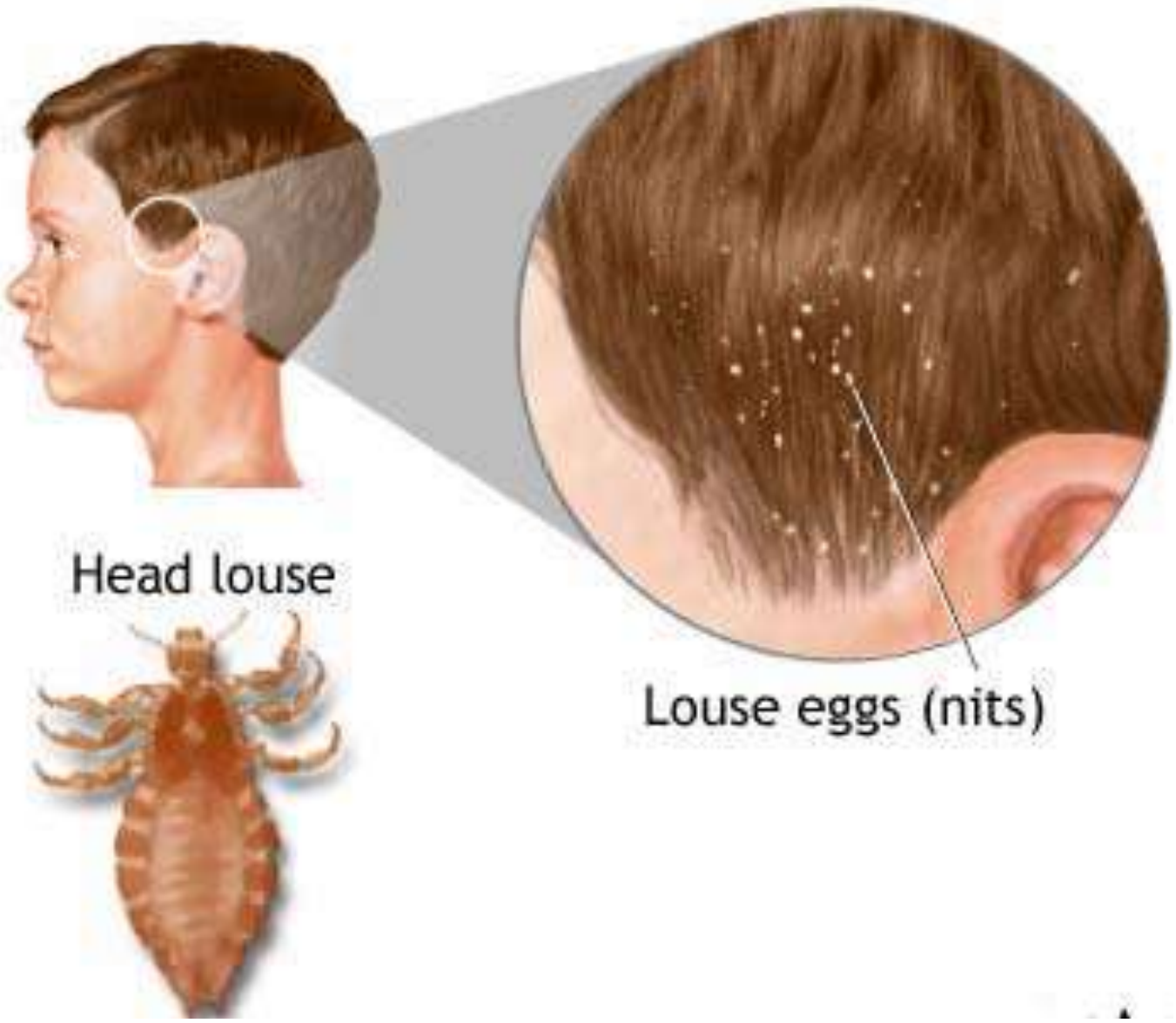


Phthiriasis pubis

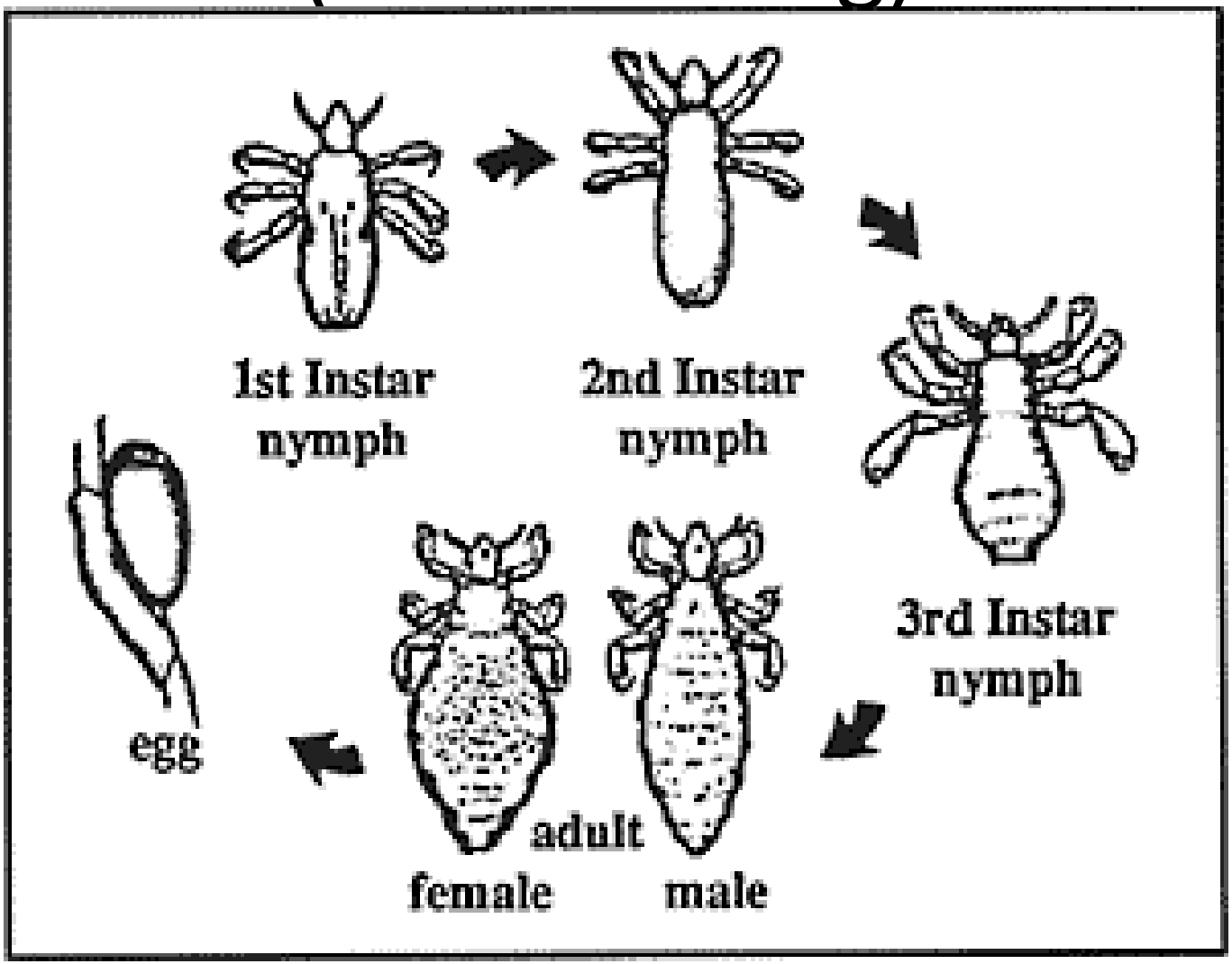


Pediculosis corporis




Head louse
Louse eggs (nits)

Pediculosis capitus life cycle (2 weeks long)



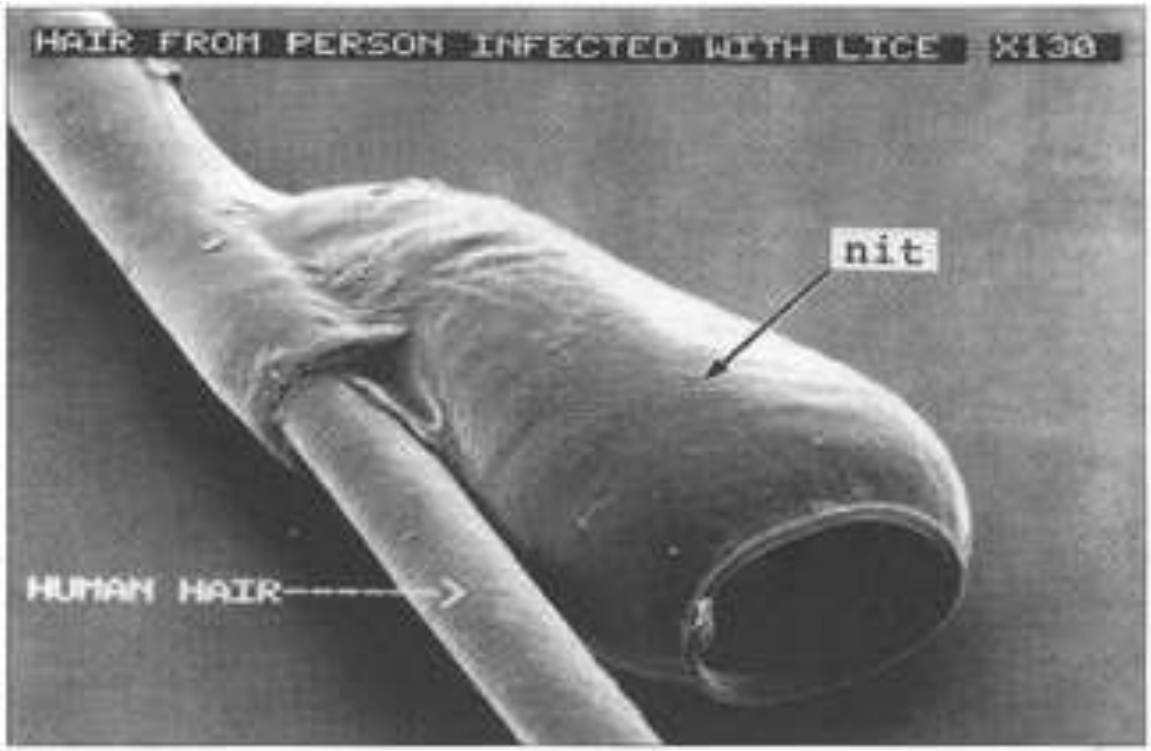
Pediculosis capitus





Pediculosis capitus





PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

Subjective

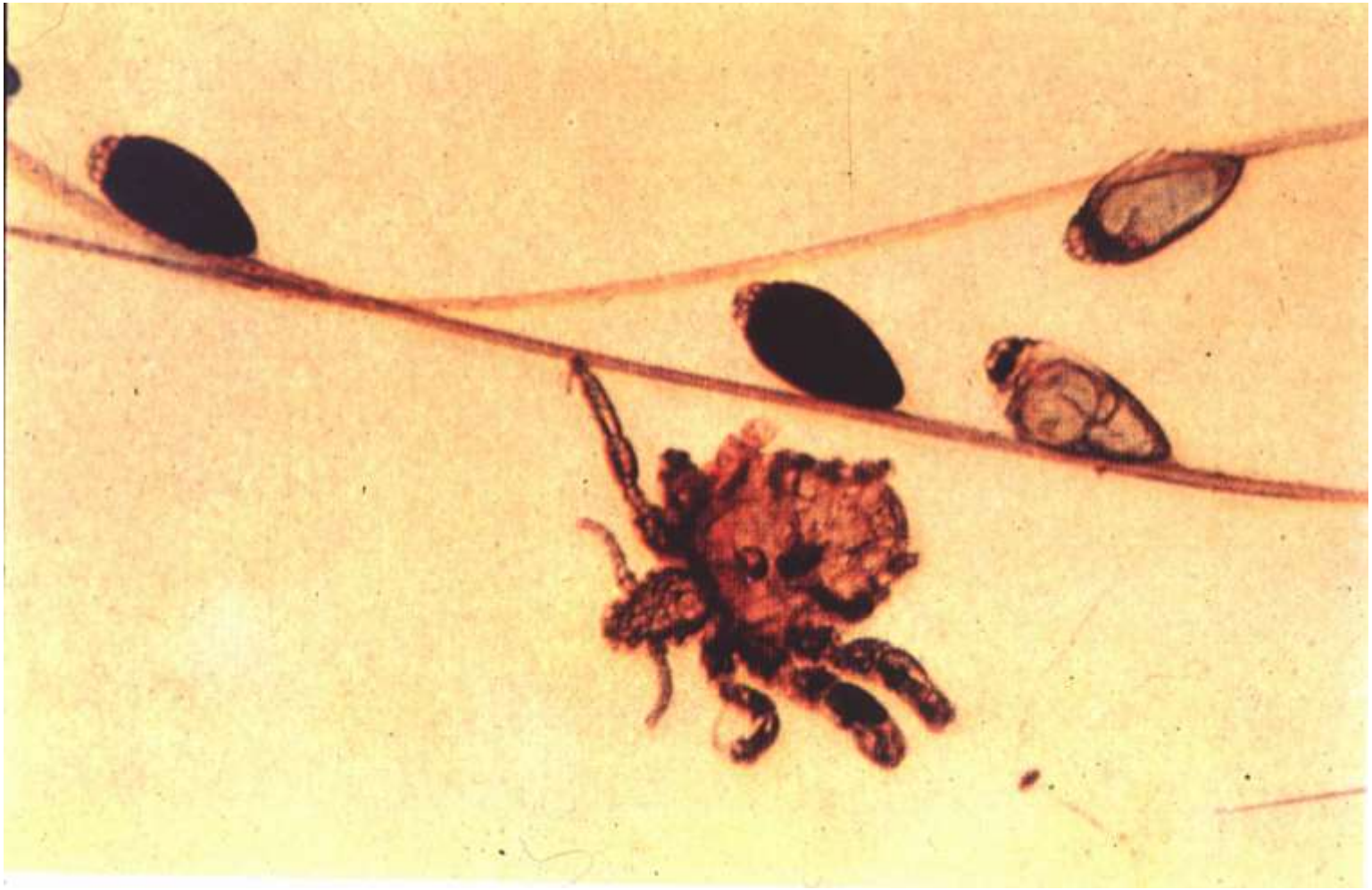
- Itching
- Red eye
- Insomnia, irritability
- Occasional preauricular lymphadenopathy

Objective

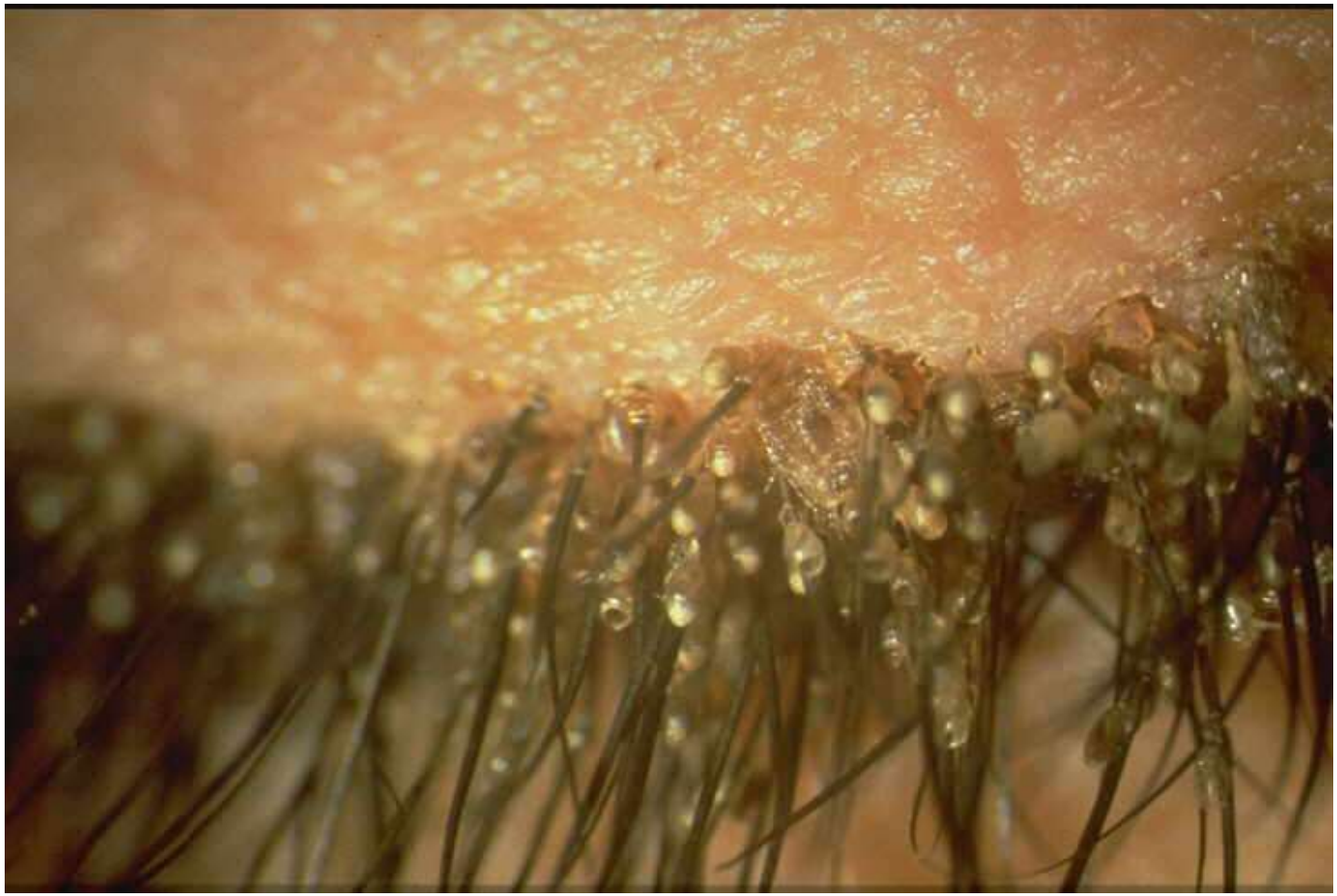
- Nit cases
- Crusty lid margins
- Madarosis
- Conj injection
- Lid edema, erythema
- Adults



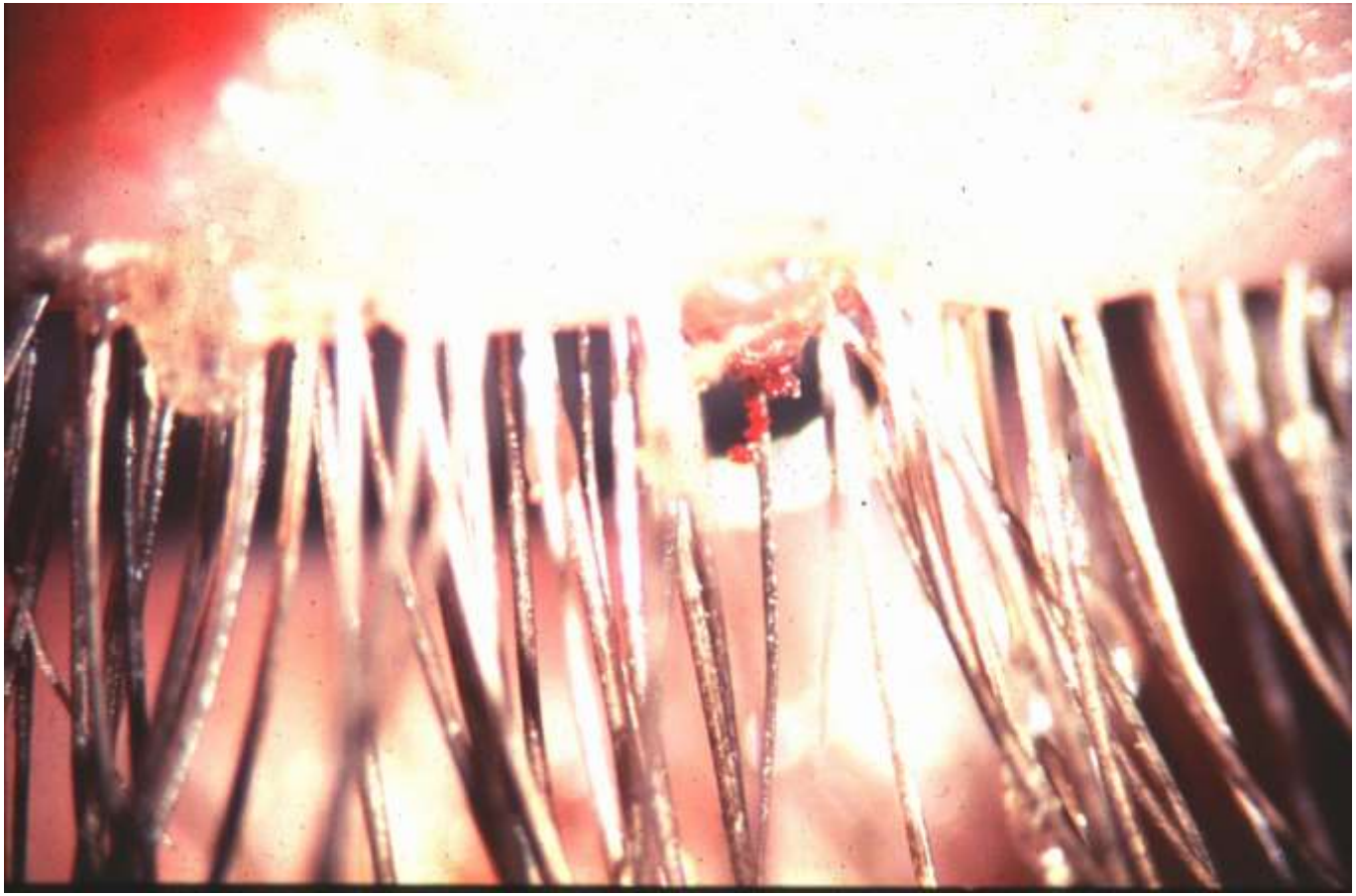
Phthiriasis



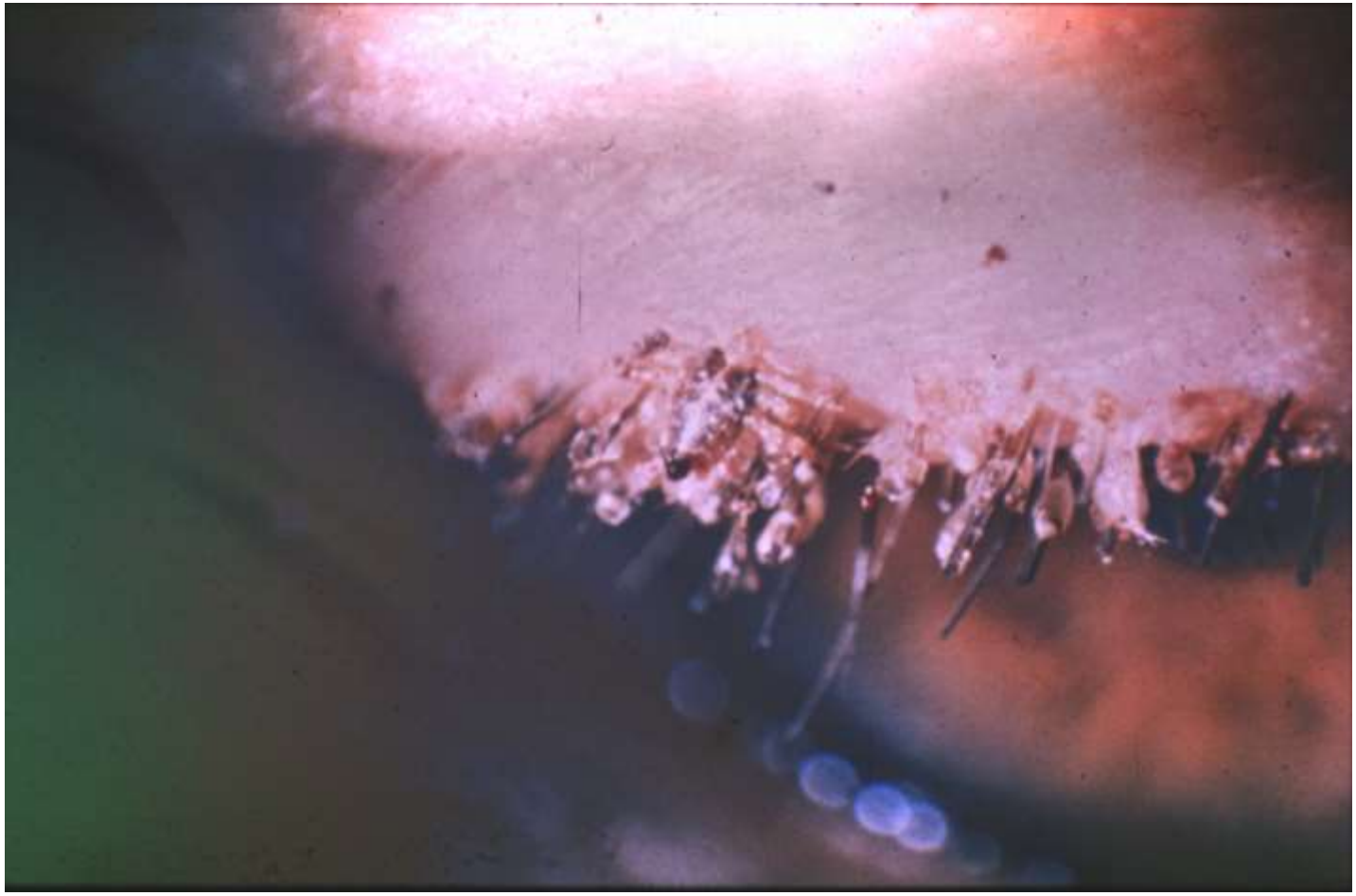
Phthiriasis pubis



Phthiriasis



Phthiriasis pubis



Phthiriasis pubis



PHTHIRIASIS PALPEBRARUM/PEDICULOSIS PALPEBRARUM aka LICE

Plan/TX

- In-office procedure (mechanical removal)
- Ung tid x 10-14 days for OCULAR tx
  - erythromycin, tobramycin, bacitracin, other ABs
  - Prasympathomimetics: physostigmine 0.25% (Eserine) ung, 2 applications 1 week apart, or pilocarpine ointment
  - Vaseline, petrolatum ointment
  - Pediculocidal agents for NON-OCULAR areas
  - Cryotherapy and argon laser have been attempted

PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

Topical anti-parastic agent options

BRAND	ACTIVE INGREDIENT	KILLS EGGS	RESISTANCE	PRICE
Nix, Rid, Pronto	Permethrin/ pyrethrin neurotoxin	No	Possibly	$10-$20
LiceMD	Dimethicone	No	No	$13-$15
Ovide RX	Malathion, neurotoxin	Yes	Not included	$125-$153
Ulesfia RX	Benzyl alcohol	No	No	$30

Management


Nix
LICE TREATMENT
With A Single Application
Kills Lice & Their Eggs
Nix


RID
LICE KILLING
SHAMPOO


RID
EGG & NIT
COMB-OUT GEL


MAXIMUM STRENGTH
PRONTO PLUS
Lice Killing
Mousse
Shampoo
Kills Lice Dead
Vitamin E
Conditions Hair


PRONTO


LiceMD
LiceMD
LiceMD


OVIDE
OVIDE


Ulesfia
(benzyl alcohol)
Lotion, 5%
Ulesfia
(benzyl alcohol)
Lotion, 5%
Sciele
Sciele



PHTHIRIASIS PALPEBRARUM/PEDICULOSIS PALPEBRARUM aka LICE

Plan/TX – less conventional

- Alcohol-coated cotton swab lid margin epithelial debridement
  - care should be taken in this procedure
- Oral ivermectin (200-400 microgram/kg as single dose; second dose option 7-10 days later)
  - Off-label (i.e. not FDA-approved), but effective
  - Head lice treatment; avoid if pregnant/age 0-5 years

PHTHIRIASIS PALPEBRARUM/ PEDICULOSIS PALPEBRARUM aka LICE

Other TX considerations:

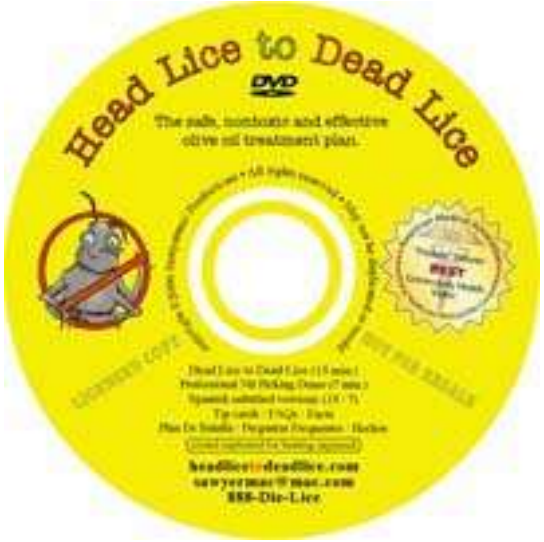
- Referral
- Family members, sexual contacts
- Home environment
- Pt. education
- Abuse issues




Let's all be
BUG
BUSTERS
All you need to know about head lice
and how to avoid them!


Fairy LiceMothers
Granting
Wishes
by Removing
Itches


HEAD LICE
BE GONE


Head Lice to Dead Lice
DVD


Bug Busting

THANK YOU!

